

Figure 8(a): High-Confidence Empirical Test Classifications (MobileNetV2 >99% Posterior Confidence)

CORRECT [TN] | Pred: Cat (99.8%)
True: Cat
Sharp triangular ear geometry & whisker patterns



CORRECT [TN] | Pred: Cat (99.5%)
True: Cat
Feline facial contour & horizontal eye alignment



CORRECT [TP] | Pred: Dog (99.8%)
True: Dog
Distinctive canine nasal bridge & elongated muzzle



CORRECT [TP] | Pred: Dog (99.6%)
True: Dog
Canine ear flop morphology & textured coat



Figure 8(b): Qualitative Failure Topology & Edge Cases (Boundary Analysis on Hard Invariant Samples)

FALSE POSITIVE [FP] | Pred: Dog (82.4%)
True: Cat
Fluffy domestic cat with flattened ears



FALSE NEGATIVE [FN] | Pred: Cat (78.6%)
True: Dog
Curled sleeping puppy with obscured muzzle



HARD OCCLUSION | Pred: Cat (65.2%)
True: Cat
Partial occlusion behind blanket boundaries



TEXTURE MIMICRY | Pred: Dog (71.8%)
True: Dog
Harsh background shadow & high contrast

